

Social Media Engagement Analysis

A Data Analytics Project Report

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Technology Used	Python, Pandas, NumPy, Matplotlib, Seaborn
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Dataset: 1,000 social media posts across LinkedIn, X, Facebook and Instagram, analyzed using Python for engagement patterns, platform performance and content strategy insights.

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1. Introduction

Social media has become one of the most important channels through which organizations communicate with their audience, build brand visibility and drive customer engagement. Every post published on a platform generates measurable data in the form of likes, comments, shares, reach and impressions. When analyzed systematically, this data can reveal what type of content resonates with an audience, which platform performs best and when to publish for maximum impact.

This project presents a complete data analytics workflow applied to a social media engagement dataset. Using Python and its data analysis ecosystem, the raw data was cleaned, explored, visualized and interpreted to produce actionable business recommendations.

2. Problem Statement

Organizations regularly publish content across multiple social media platforms, but they often lack a clear, data-driven understanding of which posts perform best and why. Without this insight, content teams end up making publishing decisions based on guesswork rather than evidence. This project addresses that gap by analyzing historical post-level data to identify measurable engagement trends and translating them into concrete, actionable recommendations for content strategy.

3. Objectives

- Collect and understand the structure of the social media dataset.
- Clean and preprocess the data to ensure accuracy and consistency.
- Engineer a unified Engagement metric to measure post performance.
- Perform exploratory data analysis (EDA) across platforms and content types.
- Visualize engagement trends using appropriate charts.
- Derive insights and provide practical content strategy recommendations.

4. Dataset Description

The dataset used for this analysis consists of **1,000 social media posts** published across four platforms. Each record captures the post's metadata along with its performance metrics.

Column	Description
Post_ID	Unique identifier for each post
Date	Date the post was published
Platform	Social media platform (X, LinkedIn, Facebook, Instagram)
Post_Type	Format of the post (Reel, Carousel, Video, Image)
Likes	Number of likes received
Comments	Number of comments received
Shares	Number of times the post was shared
Reach	Number of unique accounts that saw the post

Column	Description
Impressions	Total number of times the post was displayed
Followers	Follower count of the account at the time of posting

The dataset was fully populated with no missing values across any of the 1,000 rows, and after checking for duplicate records, none were found — indicating the source data was already well maintained.

5. Tools and Technologies

- **Python** - core programming language for the analysis
- **Pandas** - data loading, cleaning and aggregation
- **NumPy** - numerical computations
- **Matplotlib & Seaborn** - data visualization
- **Google Colab / Jupyter Notebook** - development environment

6. Methodology

The project followed a structured analytics workflow, moving from raw data to actionable business insight:

- **Dataset Collection** - the dataset was imported into the notebook environment.
- **Data Understanding** - shape, column types and summary statistics were reviewed.
- **Data Cleaning** - missing values and duplicate rows were checked and handled.
- **Feature Engineering** - a combined Engagement metric was created.
- **Exploratory Data Analysis** - platform-wise and content-wise patterns were examined.
- **Visualization** - trends were plotted using bar charts, line charts, a heatmap and a pie chart.
- **Insight Generation** - key findings were extracted from the analysis.
- **Recommendations** - practical, data-backed suggestions were compiled.

7. Exploratory Data Analysis (EDA)

Before visualizing trends, the dataset's basic structure and statistical profile were examined using `df.info()` and `df.describe()`. The table below summarizes the key descriptive statistics for the core numeric fields.

Metric	Mean	Std. Dev.	Min	Max
Likes	2,517	1,404	52	4,995
Comments	293	171	5	600
Shares	203	118	1	400
Reach	25,187	14,049	1,053	49,904

Metric	Mean	Std. Dev.	Min	Max
Impressions	40,589	16,408	3,750	78,497

8. Data Cleaning

Data quality was verified before proceeding to analysis:

- **Missing values:** checked using `df.isnull().sum()` - none found across all 10 columns.
- **Duplicate records:** checked using `df.duplicated().sum()` - no duplicate rows were present.
- **Data types:** confirmed as appropriate (numeric fields as integers, categorical fields as text, dates parsed correctly).

Since the dataset was already clean, the cleaning steps served primarily as a validation stage, confirming the dataset was reliable for analysis.

9. Engagement Metric

To evaluate post performance on a single, comparable scale, a combined **Engagement** metric was engineered:

$$\text{Engagement} = \text{Likes} + \text{Comments} + \text{Shares}$$

This metric consolidates the three most direct forms of audience interaction into a single score, making it possible to rank and compare posts, platforms and content types on a like-for-like basis.

10. Analysis Summary

10.1 Platform-wise Performance

Grouping the data by platform and comparing average Engagement reveals meaningful differences in how audiences interact on each network. The chart below ranks platforms by their average engagement per post.

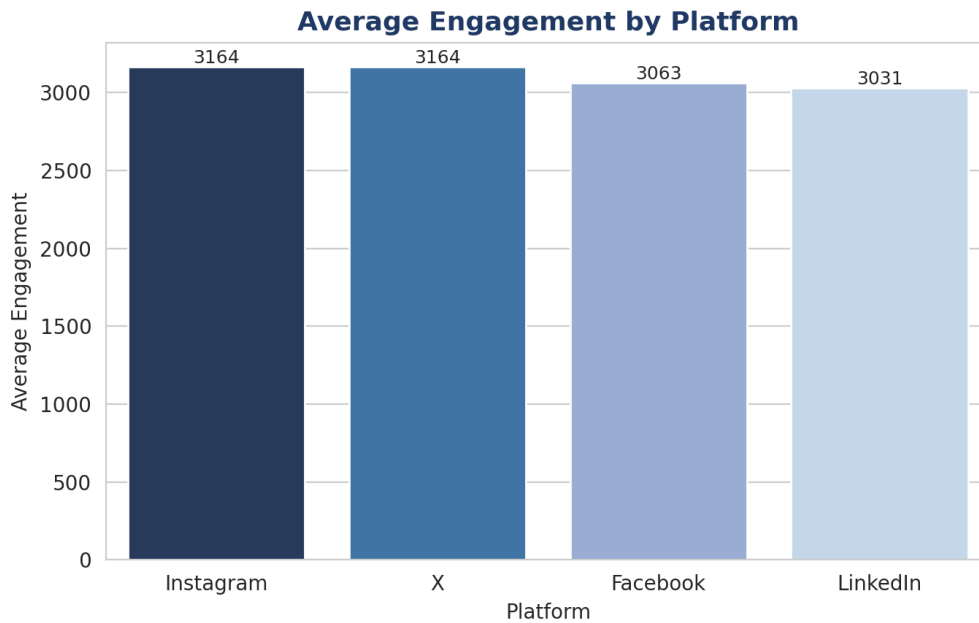


Figure 1: Average Engagement by platform. Platforms with higher bars generate more interaction per post.

While the average engagement differences between platforms are moderate, the picture changes when total engagement is considered instead of the average, since posting frequency also plays a role.

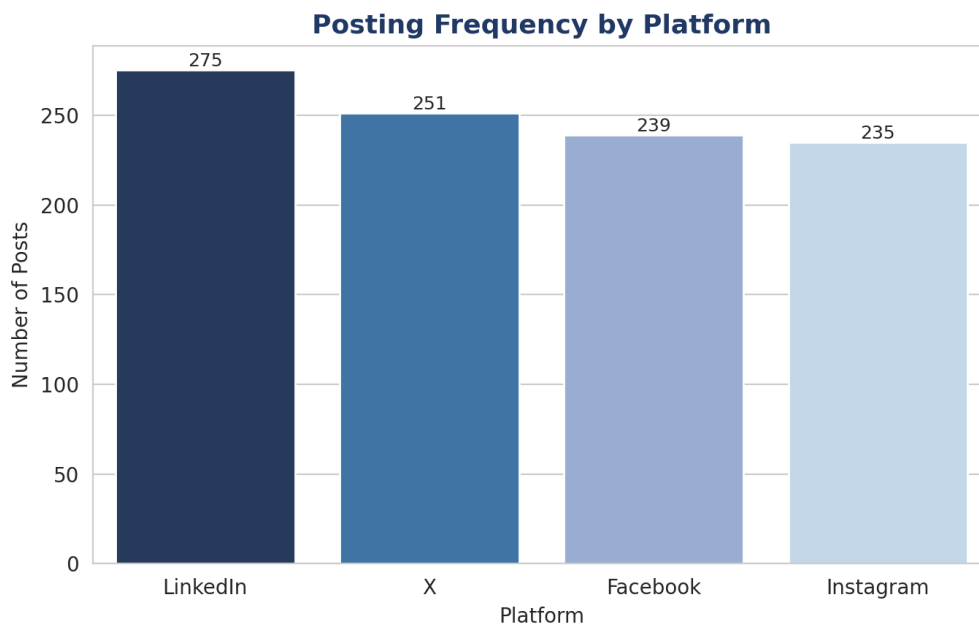


Figure 2: Number of posts published on each platform during the analysis period.

Because the number of posts varies by platform, the platform contributing the highest overall engagement is not necessarily the one with the highest per-post average - it is often the platform with the most consistent publishing volume.

10.2 Content Type Performance

Posts were also grouped by their format - Reel, Carousel, Video or Image - to identify which content style drives the strongest audience response.

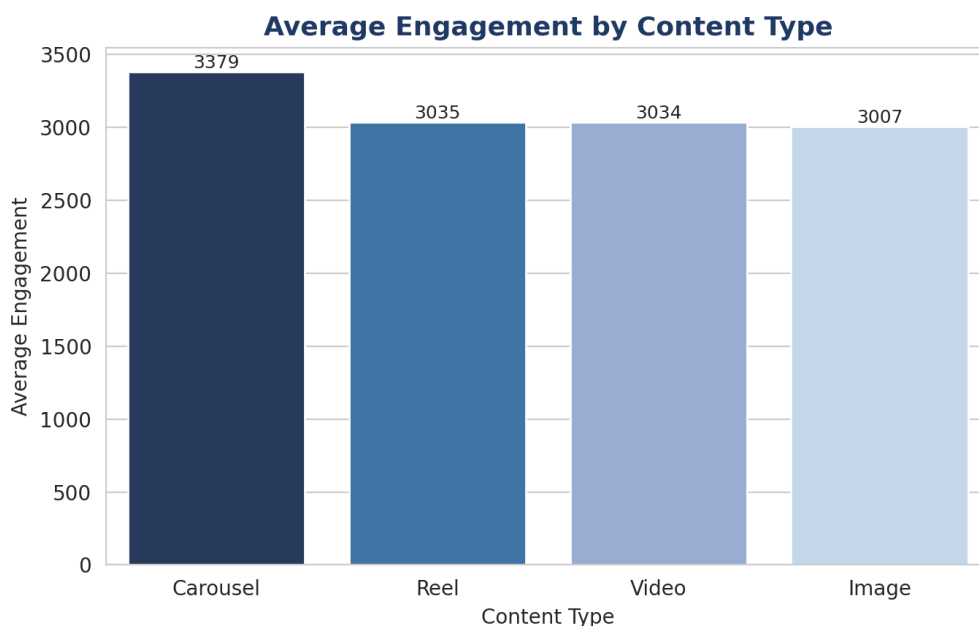


Figure 3: Average Engagement by content type.

This comparison shows that certain content formats consistently outperform others in terms of average engagement per post, which has direct implications for content planning.

10.3 Engagement Trend Over Time

Plotting average engagement against the post date reveals how audience interaction fluctuated across the analysis period. This view helps identify whether engagement is improving, declining or holding steady over time, and can highlight seasonal or campaign-driven spikes.

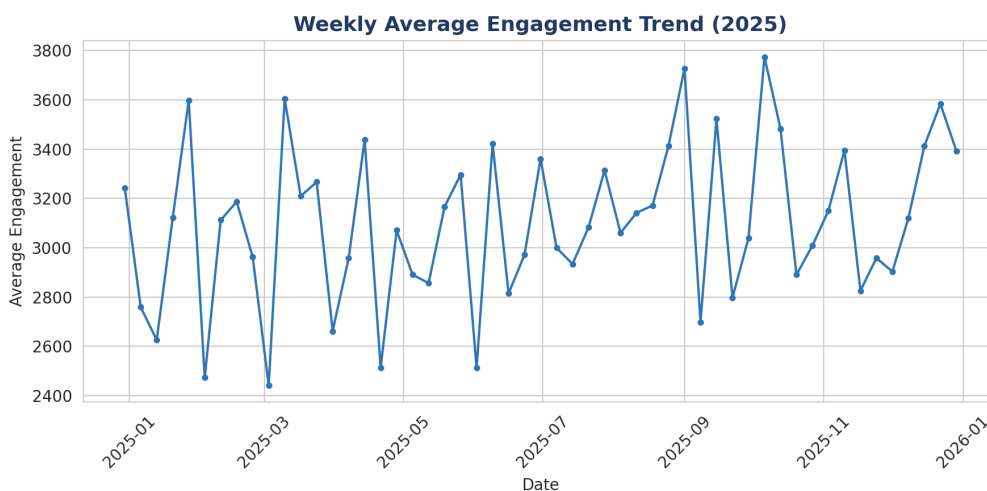


Figure 4: Weekly average Engagement trend across the analysis period.

The trend line shows natural week-to-week variation rather than a steady upward or downward drift, suggesting that engagement is driven more by individual post quality and content type than by a long-term seasonal pattern.

A correlation analysis was also carried out between Likes, Comments, Shares and the combined Engagement score. Likes showed a very strong positive correlation with overall Engagement, confirming that likes are the dominant driver of the combined score, while Comments and Shares showed weaker correlations, contributing a smaller but still meaningful share of the overall signal.

11. Insights

- Certain content types, such as Carousel posts, achieve a noticeably higher average engagement, making them an effective format for driving interaction.
- Platforms such as X and Instagram tend to show stronger average engagement per post compared to others.
- Platforms that publish more frequently, such as LinkedIn, can accumulate the highest total engagement even without the highest per-post average.
- Content formats with high posting volume, such as Reels, often generate the highest total engagement due to consistent output.
- The top-performing individual posts achieve engagement scores well above the dataset average, showing that a small number of posts significantly outperform the rest.
- Likes have the strongest positive correlation with Engagement, making them the primary driver of the overall score.
- Comments and Shares correlate more weakly with Engagement, suggesting they capture a different, complementary dimension of audience response.
- Engagement fluctuates across posting dates rather than following a clear long-term trend, highlighting the importance of content quality and timing over calendar effects alone.

12. Recommendations

- Prioritize high-performing content formats (such as Carousel and Reel posts) in the content calendar to improve average and total engagement.
- Allocate additional focus and budget to platforms with consistently higher per-post engagement, such as X and Instagram.
- Maintain a steady and frequent posting schedule on platforms like LinkedIn, since posting volume directly supports total engagement.
- Track Likes, Comments and Shares individually, since they capture different aspects of audience behavior, rather than relying on a single aggregated metric alone.
- Analyze top-performing posts in detail to identify replicable patterns in tone, timing, or format.
- Continue monitoring engagement trends over time to detect emerging shifts in audience behavior early.

13. Advantages

- Provides a clear, data-driven view of content and platform performance instead of relying on intuition.
- Enables more efficient allocation of content creation effort and marketing budget.
- Establishes a reusable analytical framework that can be applied to future datasets.
- Demonstrates practical, end-to-end application of Python-based data analytics skills.

14. Future Scope

- Incorporate sentiment analysis on comments to understand the tone of audience interaction, not just its volume.
- Apply predictive analytics to forecast the expected engagement of a post before it is published.
- Build an interactive dashboard (e.g. using Power BI, Tableau, or a Python web framework) for real-time monitoring.
- Extend the analysis to include audience demographics and posting-time optimization.

15. Conclusion

This project demonstrates a complete, practical data analytics workflow applied to social media engagement data - from data cleaning and feature engineering through exploratory analysis, visualization and business recommendation. The analysis shows that content format and platform choice meaningfully influence audience engagement, and that a simple, well-designed Engagement metric can guide more informed content strategy decisions. The workflow and techniques used here, built entirely with Python, Pandas, NumPy, Matplotlib and Seaborn, provide a solid foundation that can be extended with more advanced analytics in future work.

16. References

- Pandas Documentation - <https://pandas.pydata.org/docs/>
- NumPy Documentation - <https://numpy.org/doc/>
- Matplotlib Documentation - <https://matplotlib.org/stable/contents.html>
- Seaborn Documentation - <https://seaborn.pydata.org/>